Chapter Eighteen

Technology

Technologies

- **A technology is a process by which inputs are converted to an output.**
- ***E.g.* labor, a computer, a projector, electricity, and software are being combined to produce this lecture.**

Input Bundles

- x_i denotes the amount used of input i; *i.e.* the level of input i.
- An **input bundle** is a vector of the input levels; $(x_1, x_2, \ldots, x_n)$.
- *E.g.* $(x_1, x_2, x_3) = (6, 0, 9{\cdot}3)$.

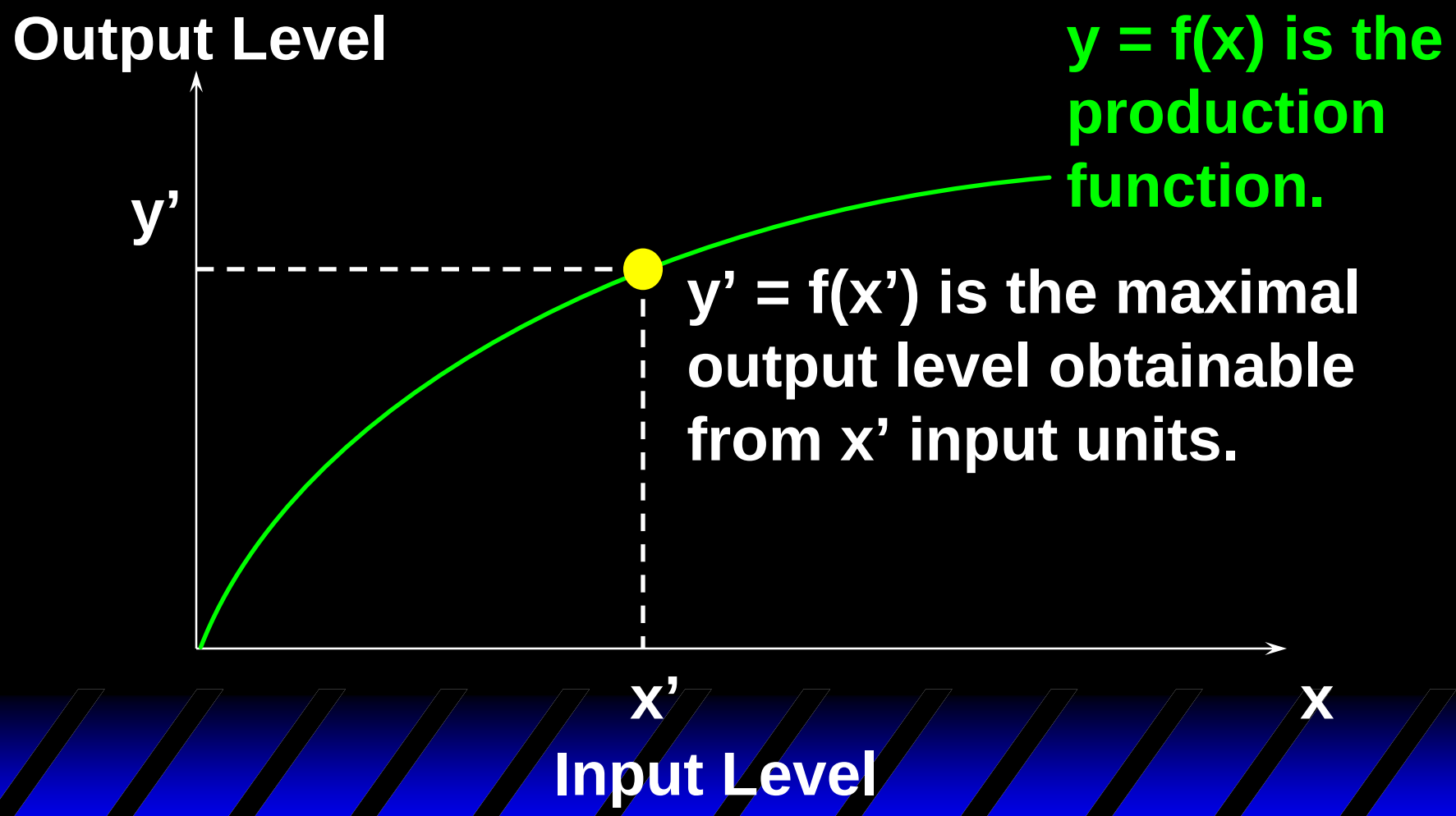
Production Functions

- y denotes the output level.
- The technology's **production function** states the **maximum** amount of output possible from an input bundle.

$$y = f(x_1, \cdots, x_n)$$

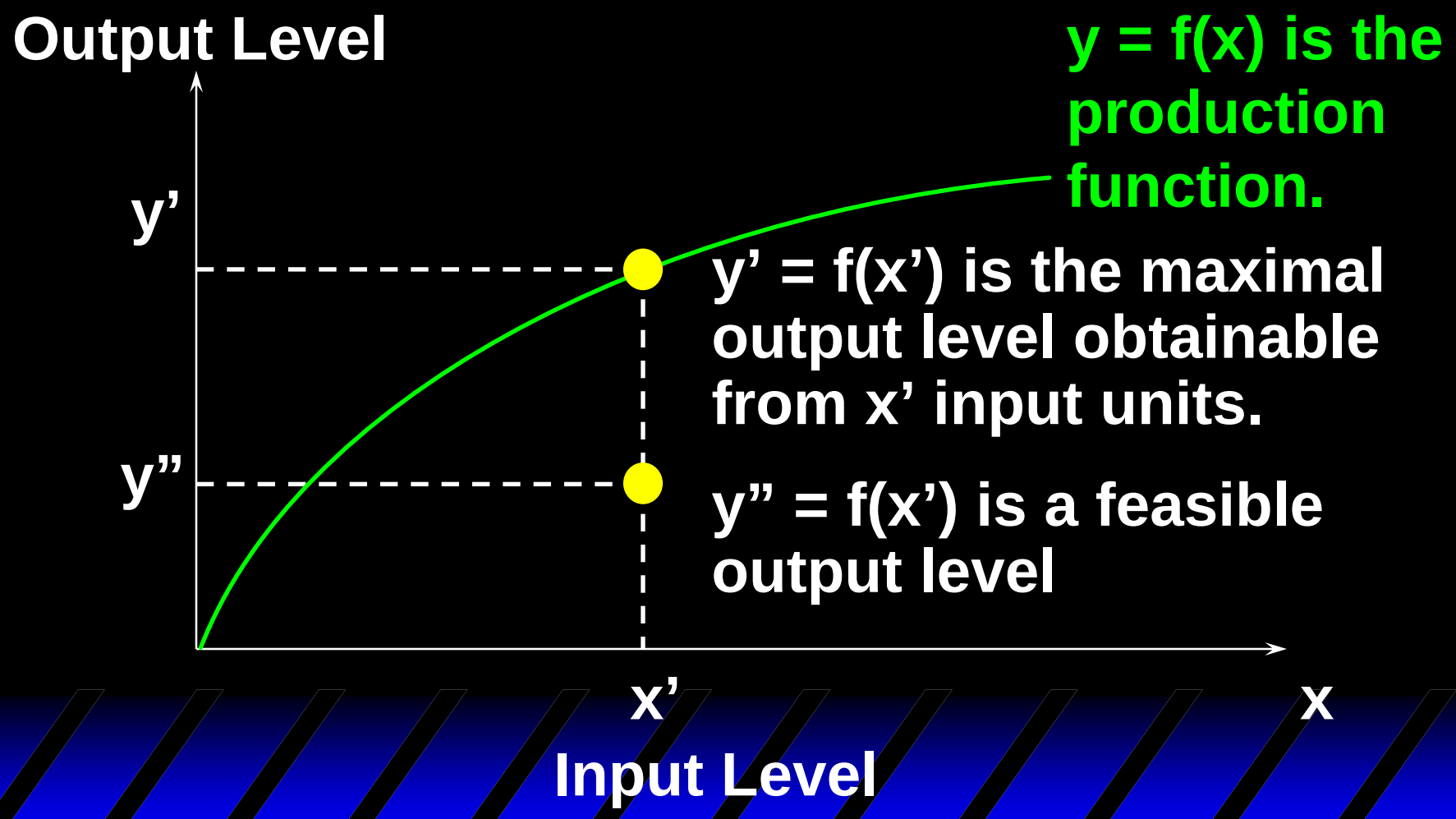
Production Functions

One input, one output



Technology Sets

One input, one output



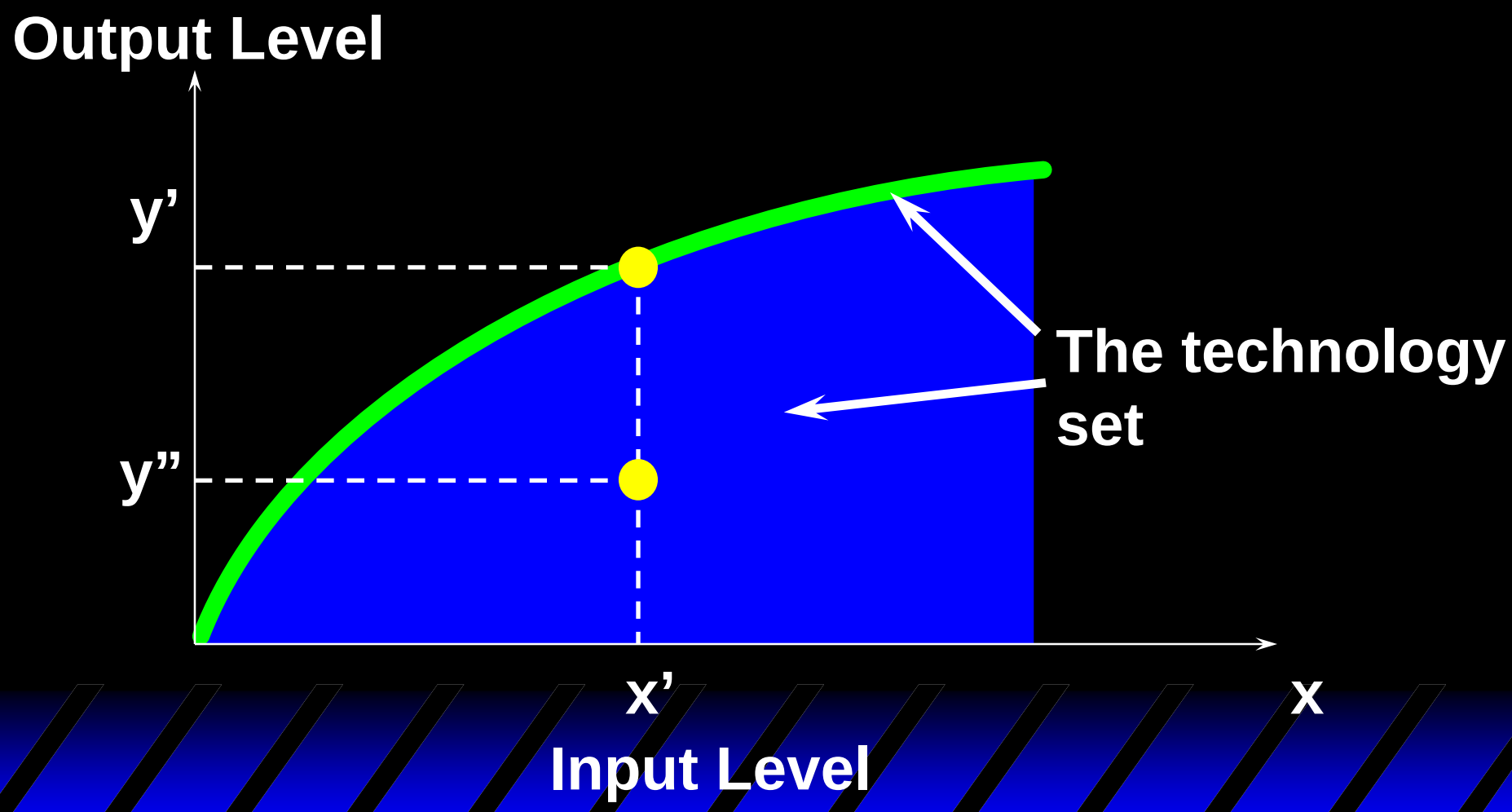
Technology Sets

The technology set is

$$T = \{(x_1, \cdots, x_n, y) \mid y \leq f(x_1, \cdots, x_n) \text{ and } x_1 \geq 0, \ldots, x_n \geq 0\}.$$

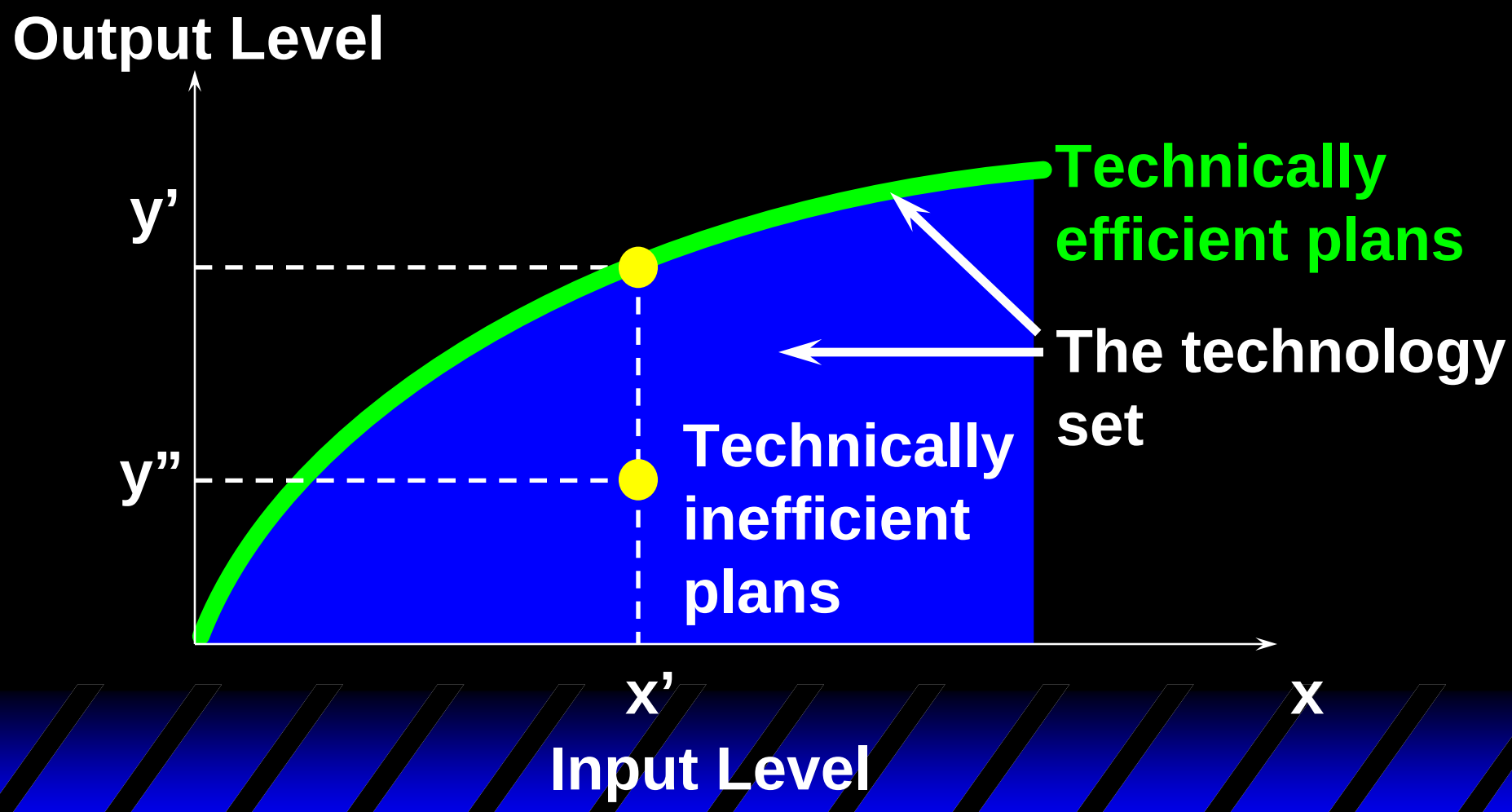
Technology Sets

One input, one output



Technology Sets

One input, one output



Technologies with Multiple Inputs

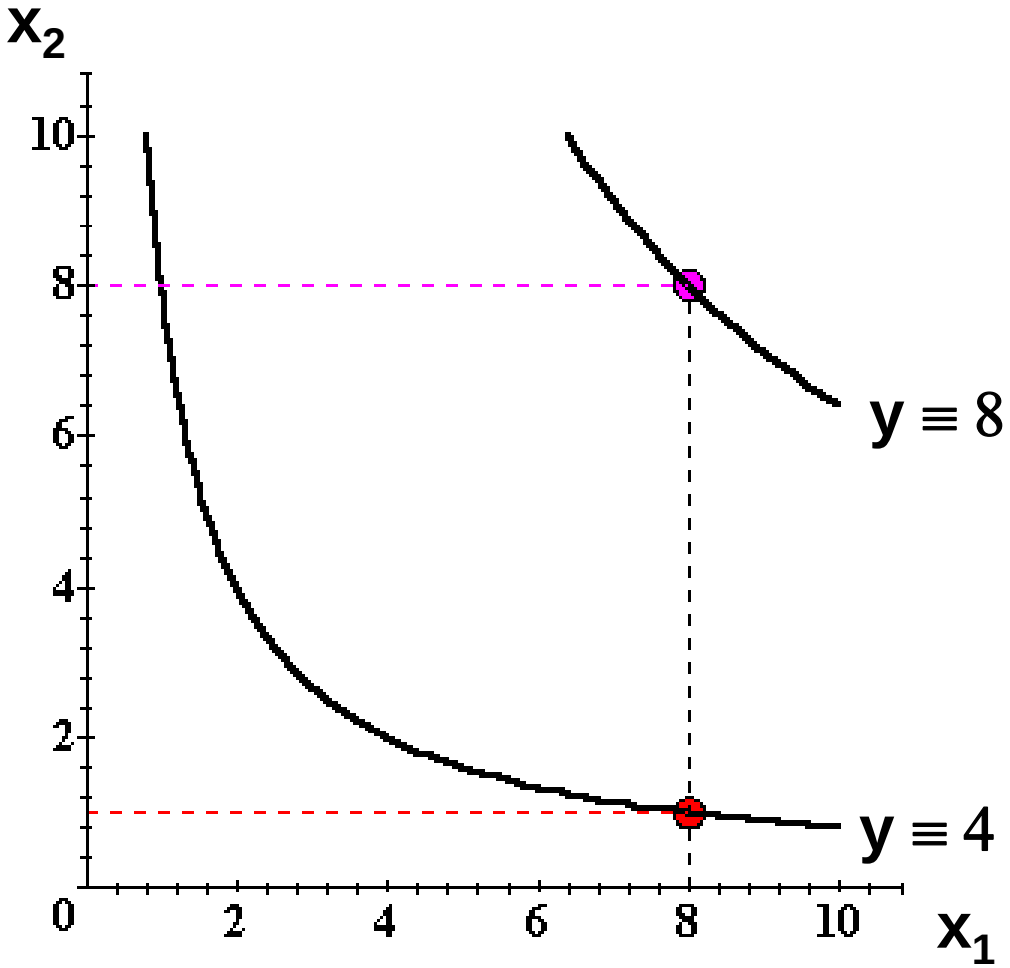
- **What does a technology look like when there is more than one input?**
- **The two input case: Input levels are x_1 and x_2. Output level is y.**
- **Example:**

$$y = f(x_1, x_2) = 2x_1^{1/3}x_2^{1/3}.$$

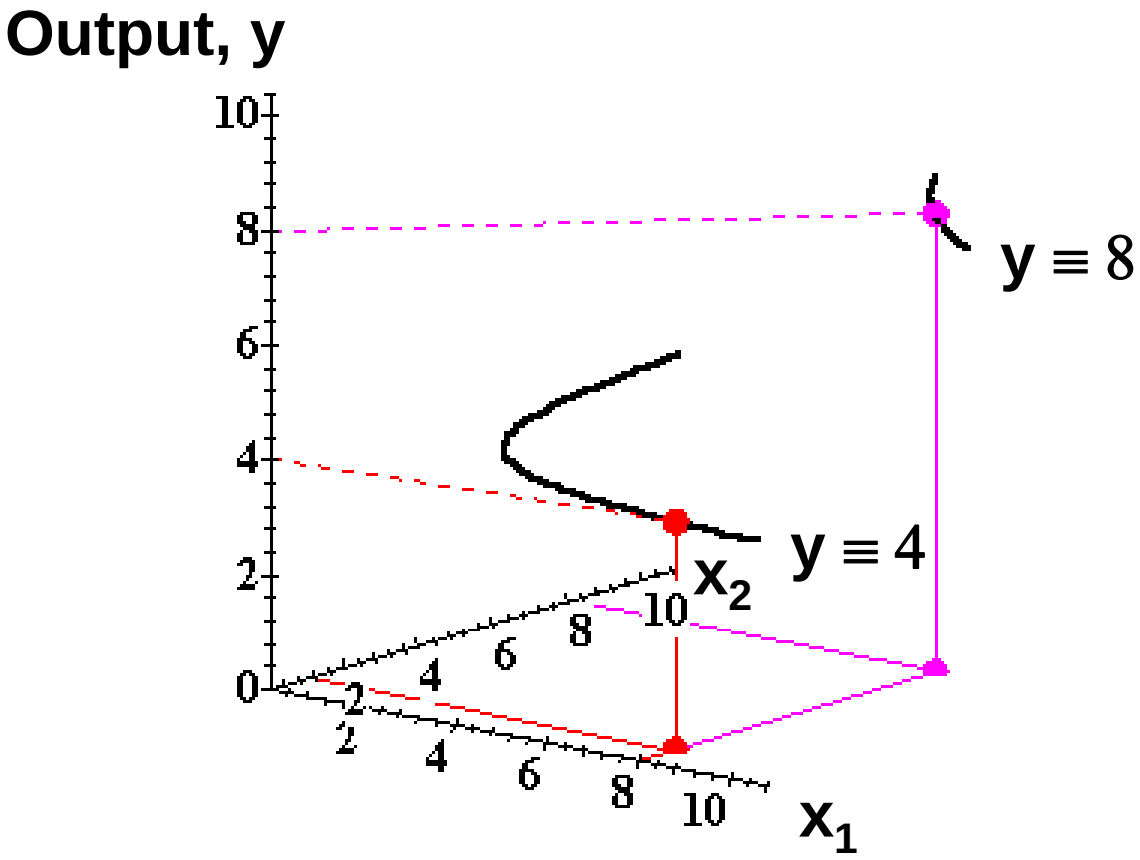
Technologies with Multiple Inputs

- **The isoquant corresponding to y units of output is the set of all input bundles that yield at most y units.**

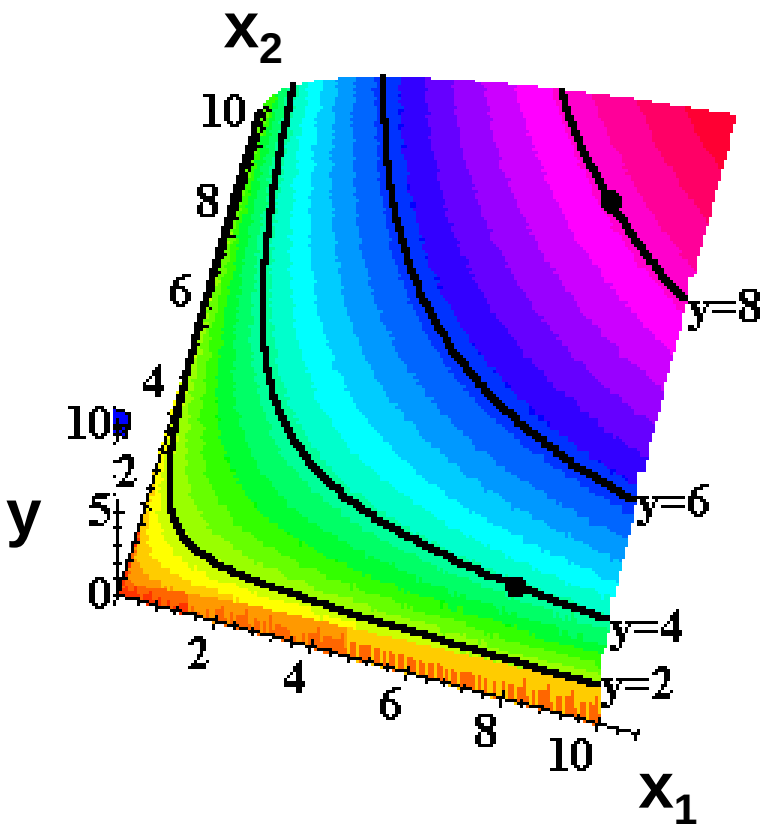
Isoquants with Two Variable Inputs



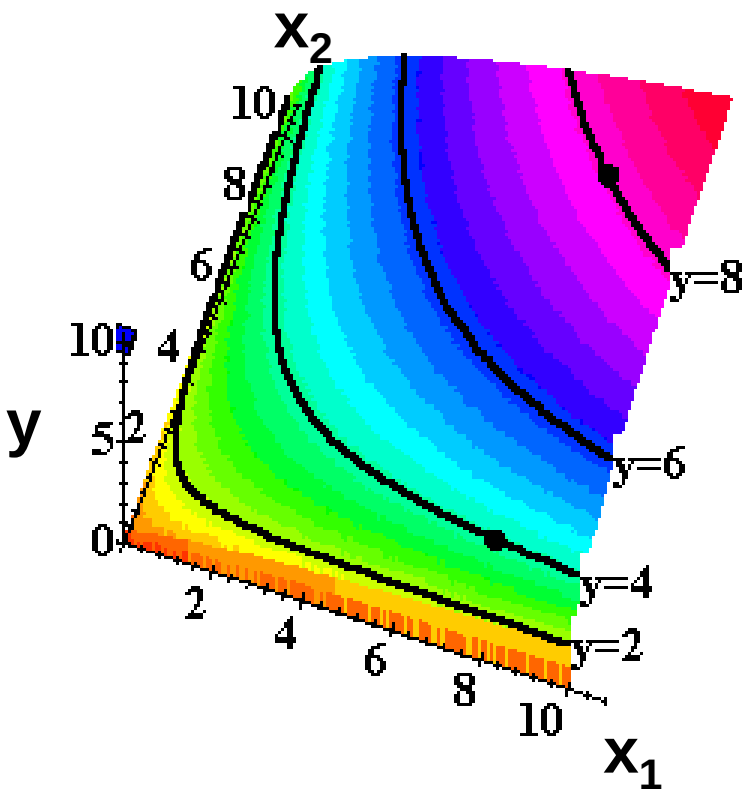
Isoquants with Two Variable Inputs



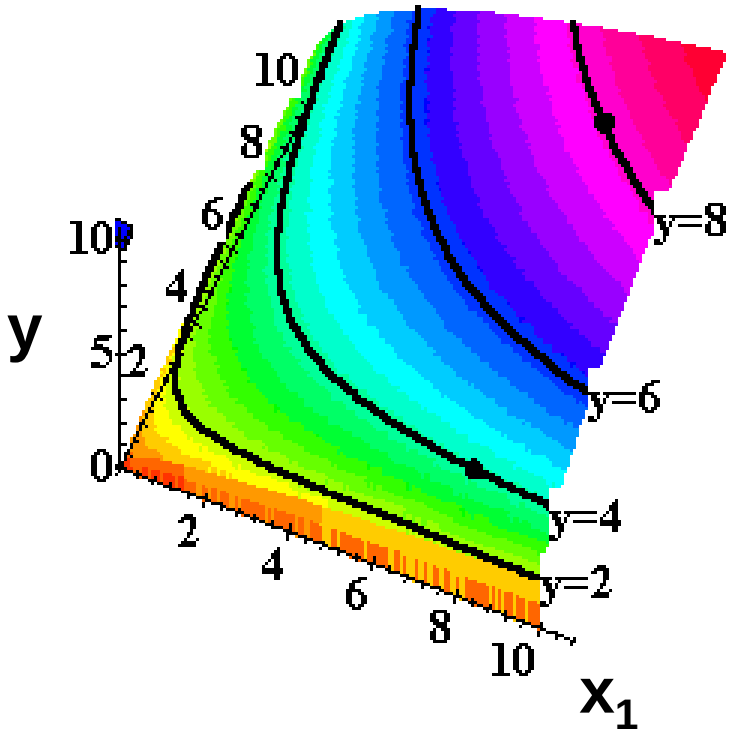
Technologies with Multiple Inputs



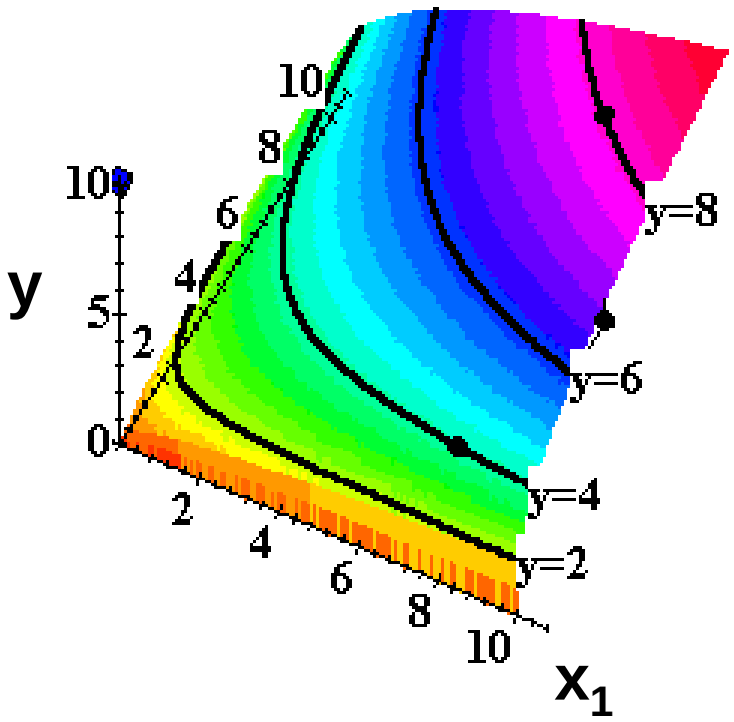
Technologies with Multiple Inputs



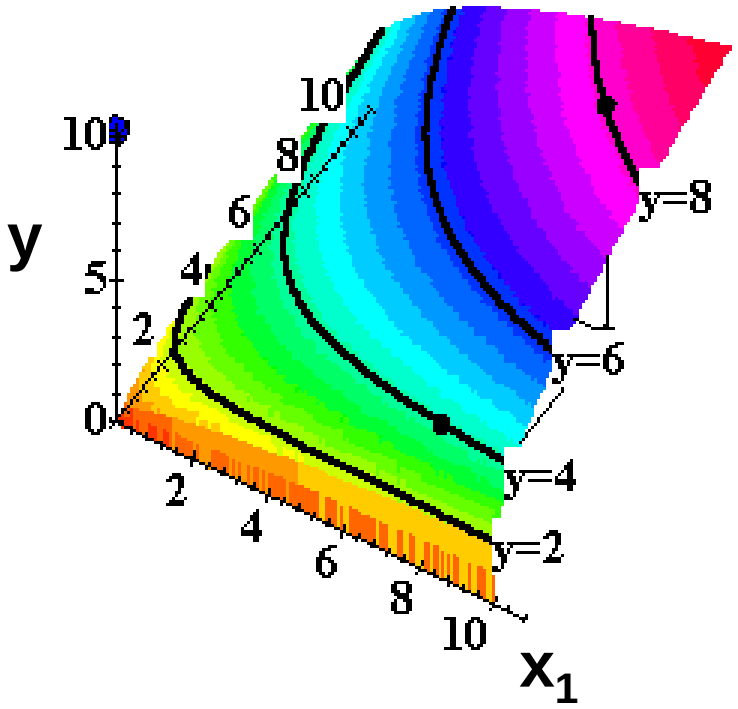
Technologies with Multiple Inputs



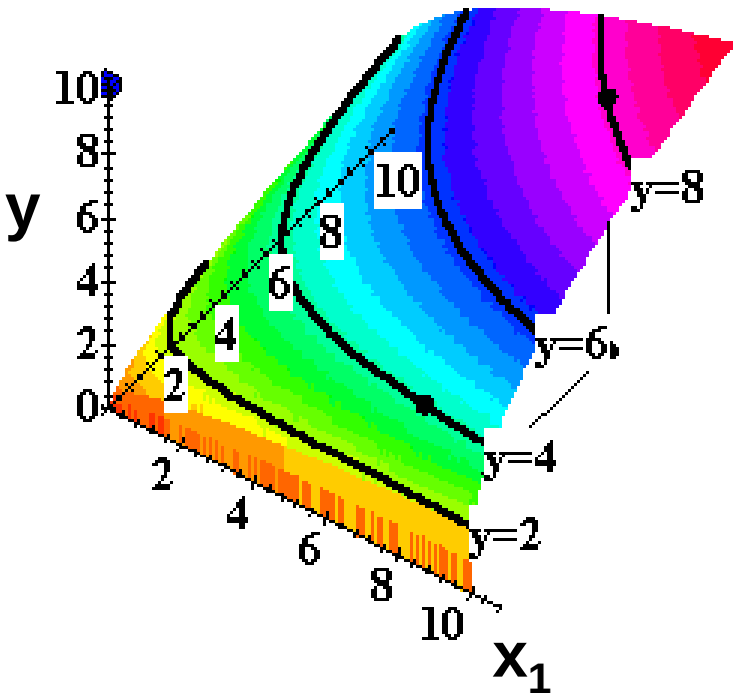
Technologies with Multiple Inputs



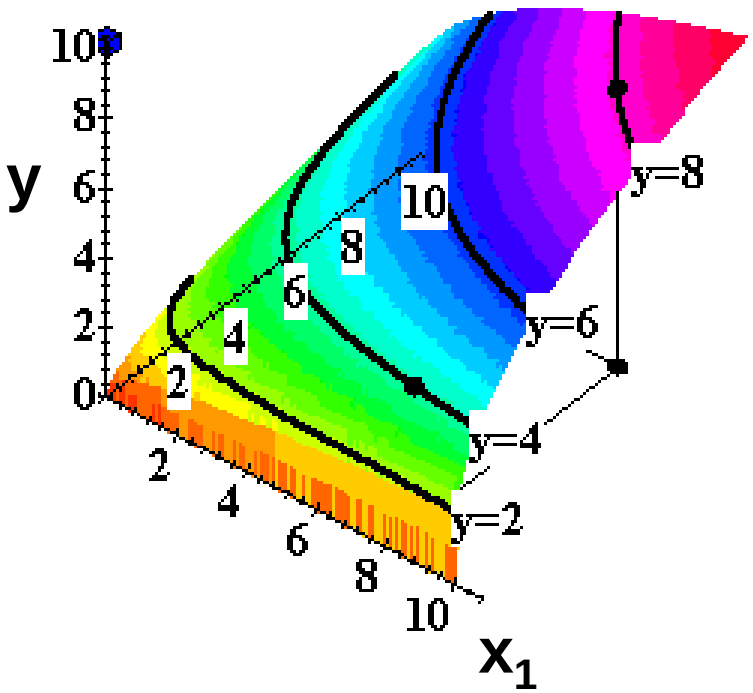
Technologies with Multiple Inputs



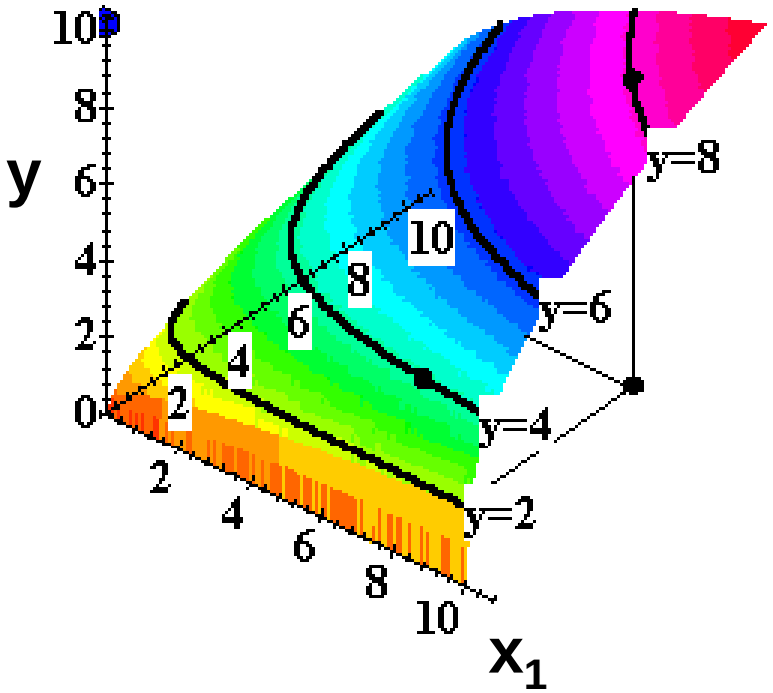
Technologies with Multiple Inputs



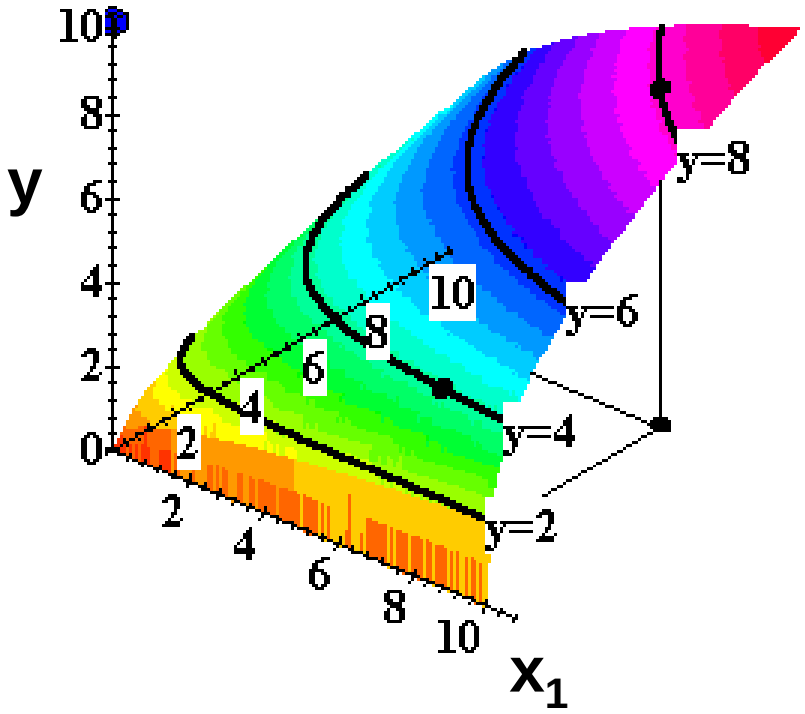
Technologies with Multiple Inputs



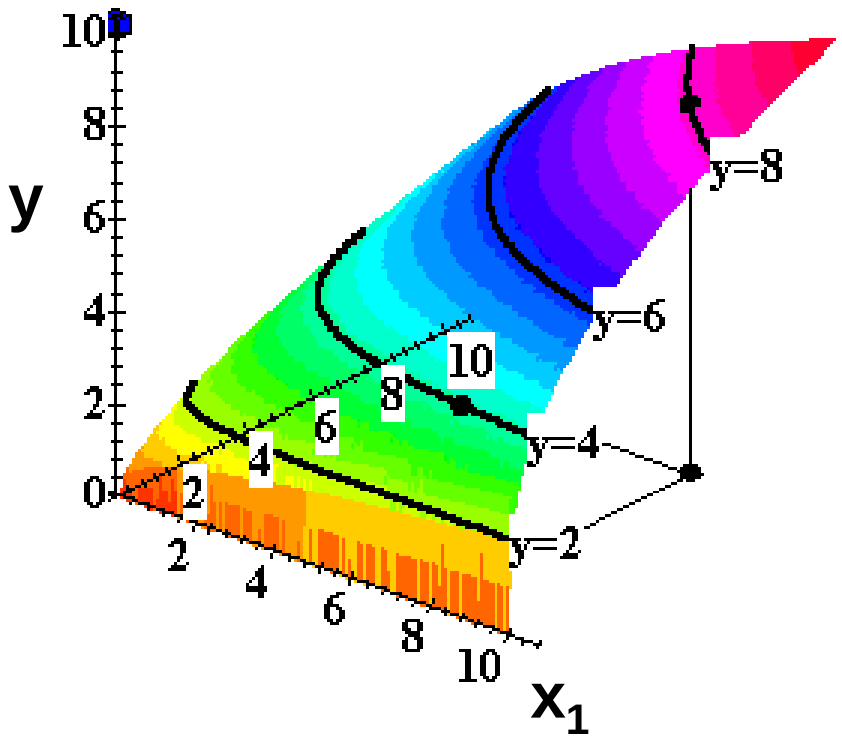
Technologies with Multiple Inputs



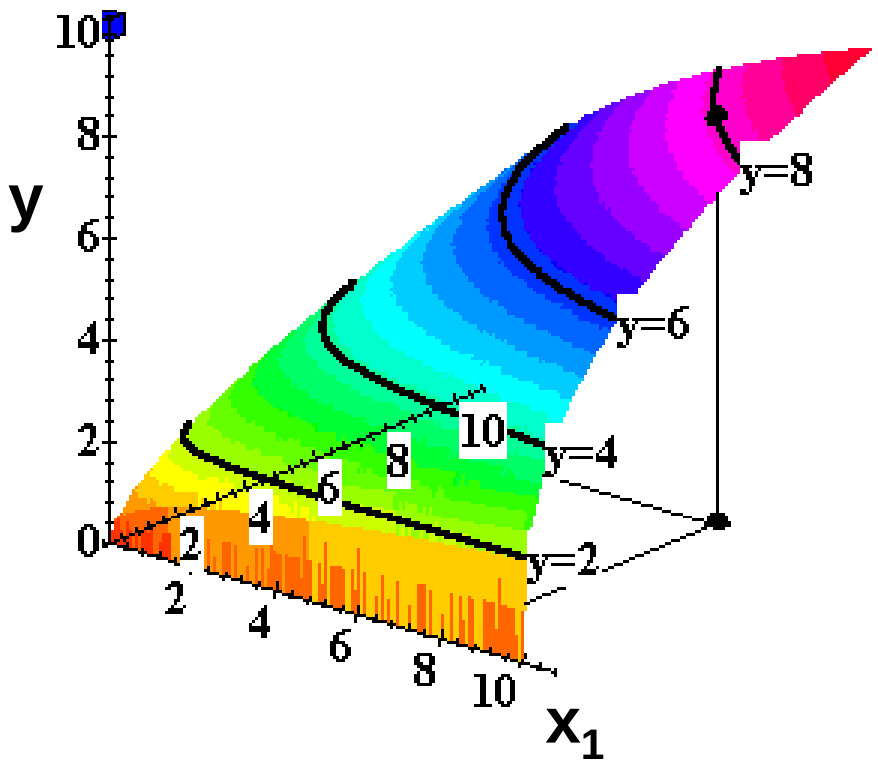
Technologies with Multiple Inputs



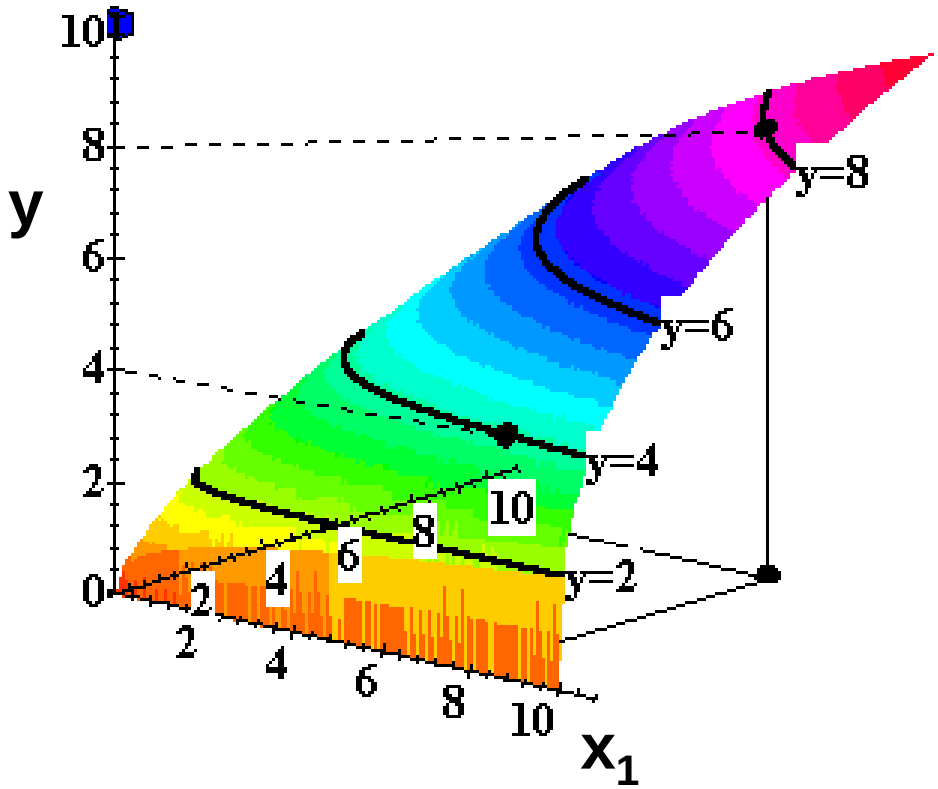
Technologies with Multiple Inputs



Technologies with Multiple Inputs



Technologies with Multiple Inputs



Marginal (Physical) Products

$$y = f(x_1, \cdots, x_n)$$

- **The marginal product of input i is the rate-of-change of the output level as the level of input i changes, holding all other input levels fixed.**
- **That is,**

$$MP_i = \frac{\partial y}{\partial x_i}$$

Marginal (Physical) Products

E.g. if

$$y = f(x_1, x_2) = x_1^{1/3} x_2^{2/3}$$

then the marginal product of input 1 is

Marginal (Physical) Products

E.g. if

$$y = f(x_1, x_2) = x_1^{1/3} x_2^{2/3}$$

then the marginal product of input 1 is

$$MP_1 = \frac{\partial y}{\partial x_1} = \frac{1}{3} x_1^{-2/3} x_2^{2/3}$$

Marginal (Physical) Products

E.g. if

$$y = f(x_1, x_2) = x_1^{1/3} x_2^{2/3}$$

then the marginal product of input 1 is

$$MP_1 = \frac{\partial y}{\partial x_1} = \frac{1}{3} x_1^{-2/3} x_2^{2/3}$$

and the marginal product of input 2 is

Marginal (Physical) Products

E.g. if

$$y = f(x_1, x_2) = x_1^{1/3} x_2^{2/3}$$

then the marginal product of input 1 is

$$MP_1 = \frac{\partial y}{\partial x_1} = \frac{1}{3} x_1^{-2/3} x_2^{2/3}$$

and the marginal product of input 2 is

$$MP_2 = \frac{\partial y}{\partial x_2} = \frac{2}{3} x_1^{1/3} x_2^{-1/3}.$$

Marginal (Physical) Products

- **The marginal product of input i is diminishing if it becomes smaller as the level of input i increases. That is, if**

$$\frac{\partial MP_i}{\partial x_i} = \frac{\partial}{\partial x_i}\left(\frac{\partial y}{\partial x_i}\right) = \frac{\partial^2 y}{\partial x_i^2} < 0.$$

- **True in the Cobb-Douglas example.**

Returns-to-Scale

- **Marginal products describe the change in output level as a single input level changes.**
- **Returns-to-scale describes how the output level changes as all input levels change in direct proportion (*e.g.* all input levels doubled)**

Returns-to-Scale

If, for any input bundle $(x_1,\ldots,x_n)$,

$$f(kx_1,kx_2,\cdots,kx_n) = kf(x_1,x_2,\cdots,x_n)$$

then the technology described by the production function f exhibits constant returns-to-scale.

***E.g*. (k = 2) doubling all input levels doubles the output level.**

Returns-to-Scale

If, for any input bundle $(x_1,\ldots,x_n)$,

$$f(kx_1, kx_2, \cdots, kx_n) < kf(x_1, x_2, \cdots, x_n)$$

then the technology exhibits diminishing returns-to-scale.

***E.g*. (k = 2) doubling all input levels less than doubles the output level.**

Returns-to-Scale

If, for any input bundle $(x_1,\ldots,x_n)$,

$$f(kx_1, kx_2, \cdots, kx_n) > kf(x_1, x_2, \cdots, x_n)$$

then the technology exhibits increasing returns-to-scale.

***E.g*. (k = 2) doubling all input levels more than doubles the output level.**

Examples of Returns-to-Scale

The Cobb-Douglas production function is

$$y = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}.$$

Expand all input levels proportionately by k. The output level becomes

$$(kx_1)^{a_1}(kx_2)^{a_2}\cdots(kx_n)^{a_n}$$

Examples of Returns-to-Scale

The Cobb-Douglas production function is

$$y = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}.$$

Expand all input levels proportionately by k. The output level becomes

$$(kx_1)^{a_1}(kx_2)^{a_2}\cdots(kx_n)^{a_n}$$

$$= k^{a_1}k^{a_2}\cdots k^{a_n}x^{a_1}x^{a_2}\cdots x^{a_n}$$

Examples of Returns-to-Scale

The Cobb-Douglas production function is

$$y = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}.$$

Expand all input levels proportionately by k. The output level becomes

$$(kx_1)^{a_1}(kx_2)^{a_2}\cdots(kx_n)^{a_n}$$

$$= k^{a_1}k^{a_2}\cdots k^{a_n}x^{a_1}x^{a_2}\cdots x^{a_n}$$

$$= k^{a_1+a_2+\cdots+a_n}x_1^{a_1}x_2^{a_2}\cdots x_n^{a_n}$$

Examples of Returns-to-Scale

The Cobb-Douglas production function is

$$y = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}.$$

Expand all input levels proportionately by k. The output level becomes

$$(kx_1)^{a_1}(kx_2)^{a_2}\cdots(kx_n)^{a_n}$$

$$= k^{a_1}k^{a_2}\cdots k^{a_n}x^{a_1}x^{a_2}\cdots x^{a_n}$$

$$= k^{a_1+a_2+\cdots+a_n}x_1^{a_1}x_2^{a_2}\cdots x_n^{a_n}$$

$$= k^{a_1+\cdots+a_n}y.$$

Examples of Returns-to-Scale

The Cobb-Douglas production function is

$$y = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}.$$

$$(kx_1)^{a_1}(kx_2)^{a_2}\cdots(kx_n)^{a_n} = k^{a_1+\cdots+a_n} y.$$

The Cobb-Douglas technology's returns-to-scale is

constant **if** $a_1 + \ldots + a_n = 1$

Examples of Returns-to-Scale

The Cobb-Douglas production function is

$$y = x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n}.$$

$$(kx_1)^{a_1}(kx_2)^{a_2}\cdots(kx_n)^{a_n} = k^{a_1+\cdots+a_n} y.$$

The Cobb-Douglas technology's returns-to-scale is

constant if $a_1 + \ldots + a_n = 1$

increasing if $a_1 + \ldots + a_n > 1$

Examples of Returns-to-Scale

The Cobb-Douglas technology's returns-to-scale is

constant if $a_1 + \ldots + a_n = 1$

increasing if $a_1 + \ldots + a_n > 1$

decreasing if $a_1 + \ldots + a_n < 1$.

Returns-to-Scale

- **Q: Can a technology exhibit increasing returns-to-scale even though all of its marginal products are diminishing?**

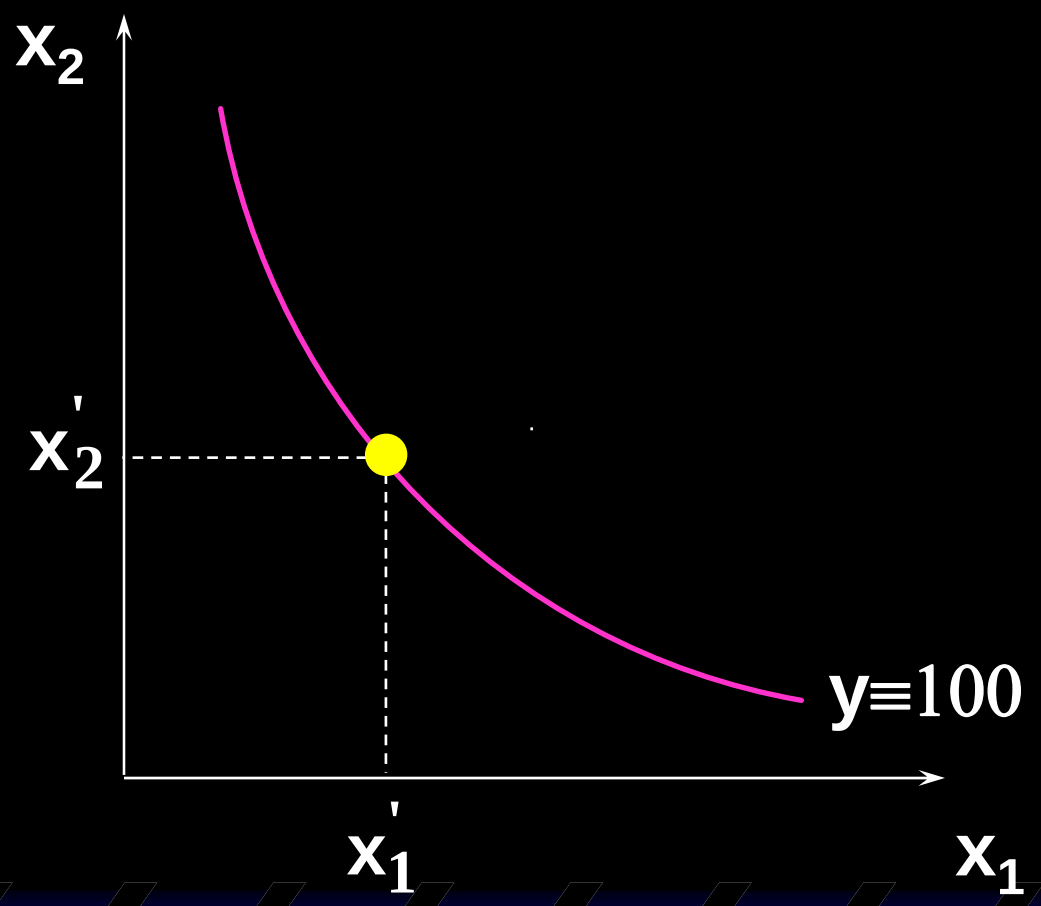
Returns-to-Scale

- **Q: Can a technology exhibit increasing returns-to-scale even if all of its marginal products are diminishing?**
- **A: Yes.**
- **E.g.** $y = x_1^{2/3} x_2^{2/3}$.

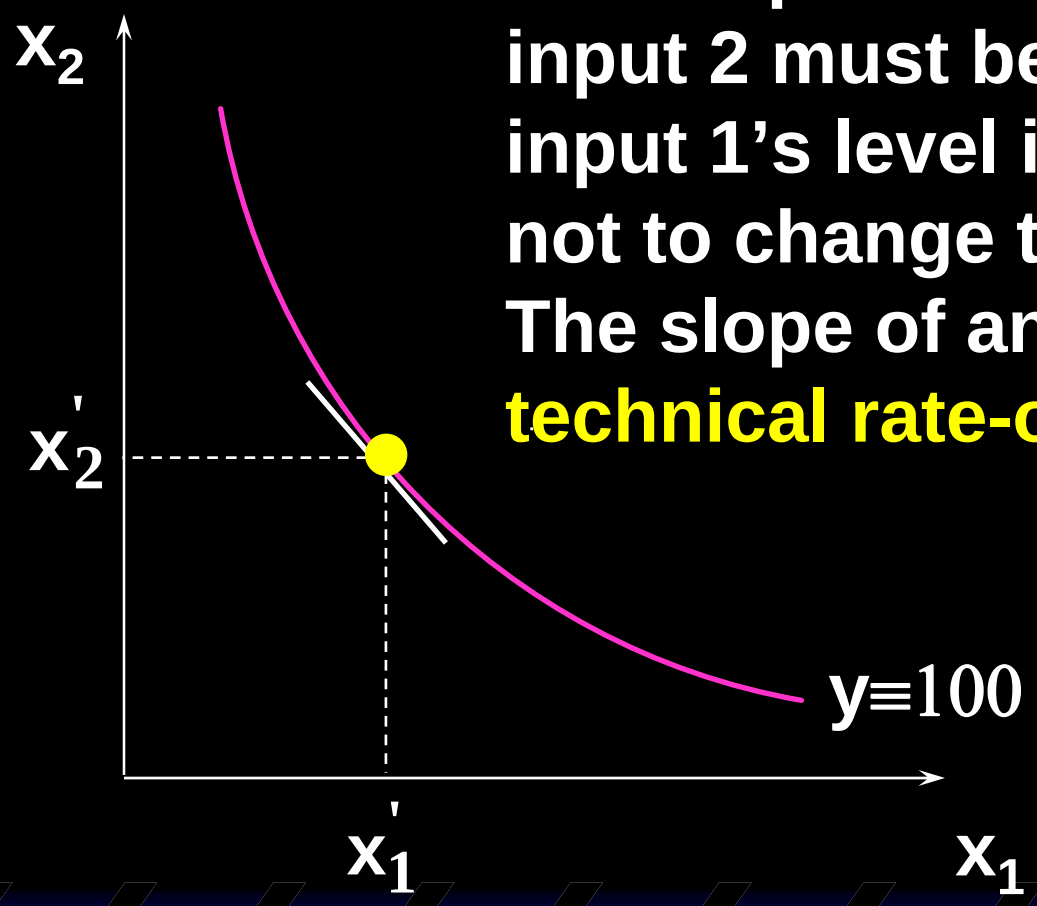
Technical Rate-of-Substitution

- **At what rate can a firm substitute one input for another without changing its output level?**

Technical Rate-of-Substitution



Technical Rate-of-Substitution



The slope is the rate at which input 2 must be given up as input 1's level is increased so as not to change the output level. The slope of an isoquant is its technical rate-of-substitution.

Technical Rate-of-Substitution

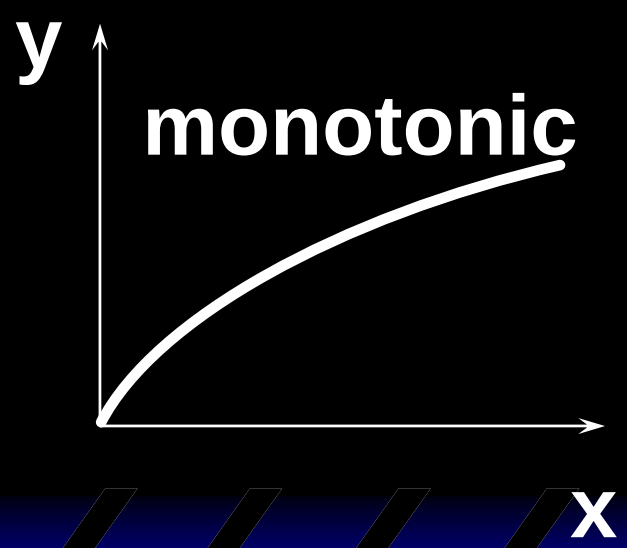
$$\frac{dx_2}{dx_1} = -\frac{\partial y / \partial x_1}{\partial y / \partial x_2}$$

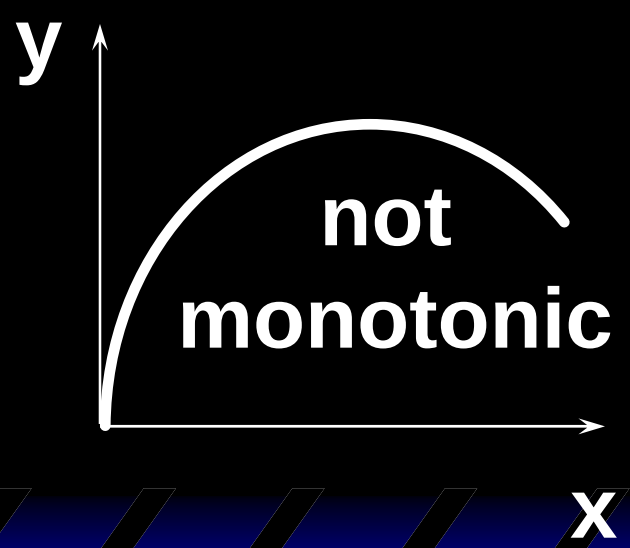
Well-Behaved Technologies

- A **well-behaved** technology is
  - **monotonic**, and
  - **convex**.

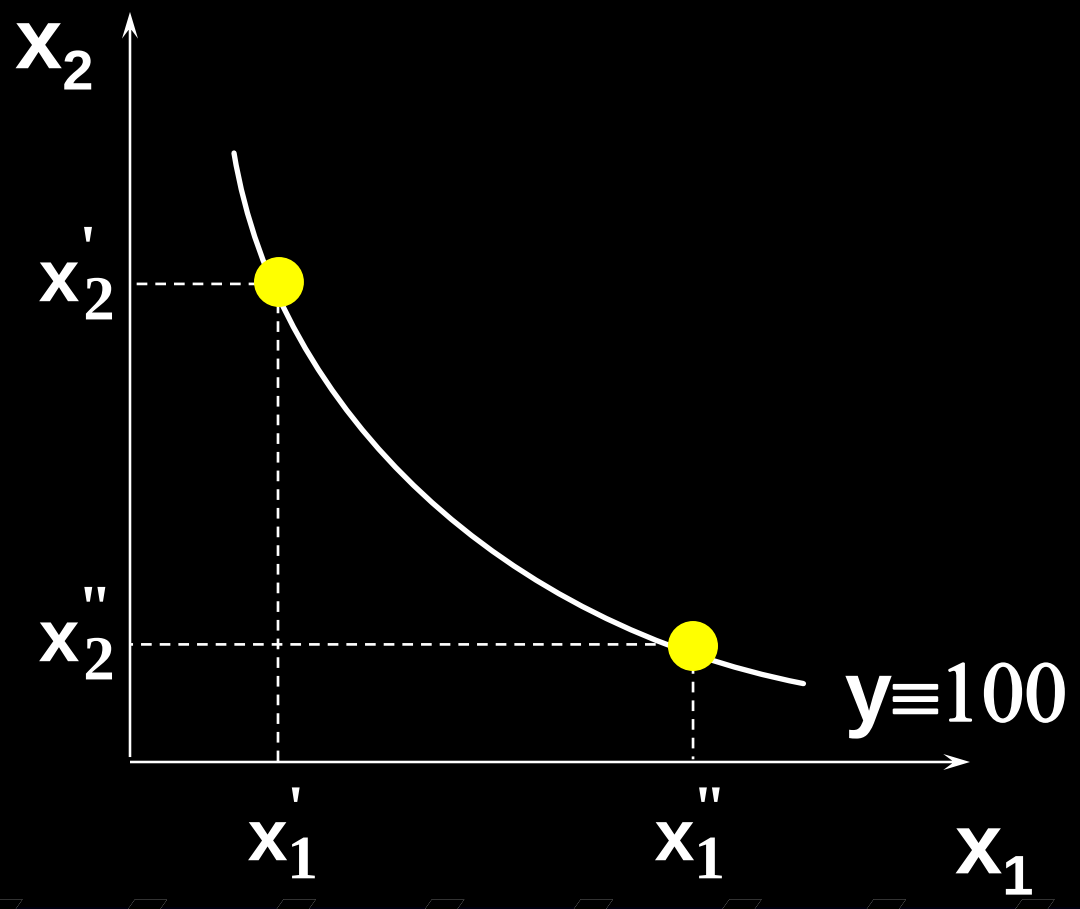
Well-Behaved Technologies - Monotonicity

- **Monotonicity: More of any input generates more output.**

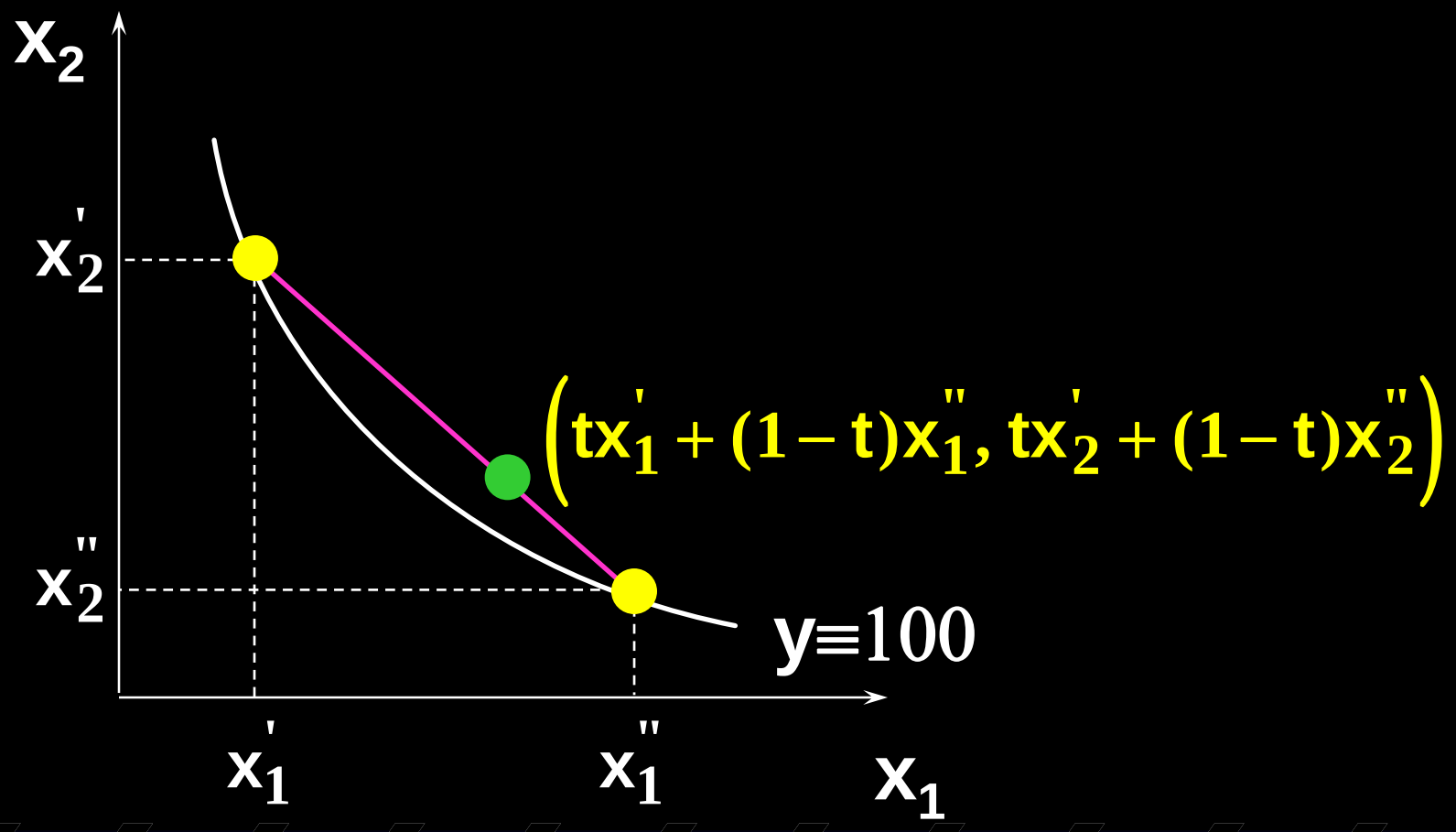




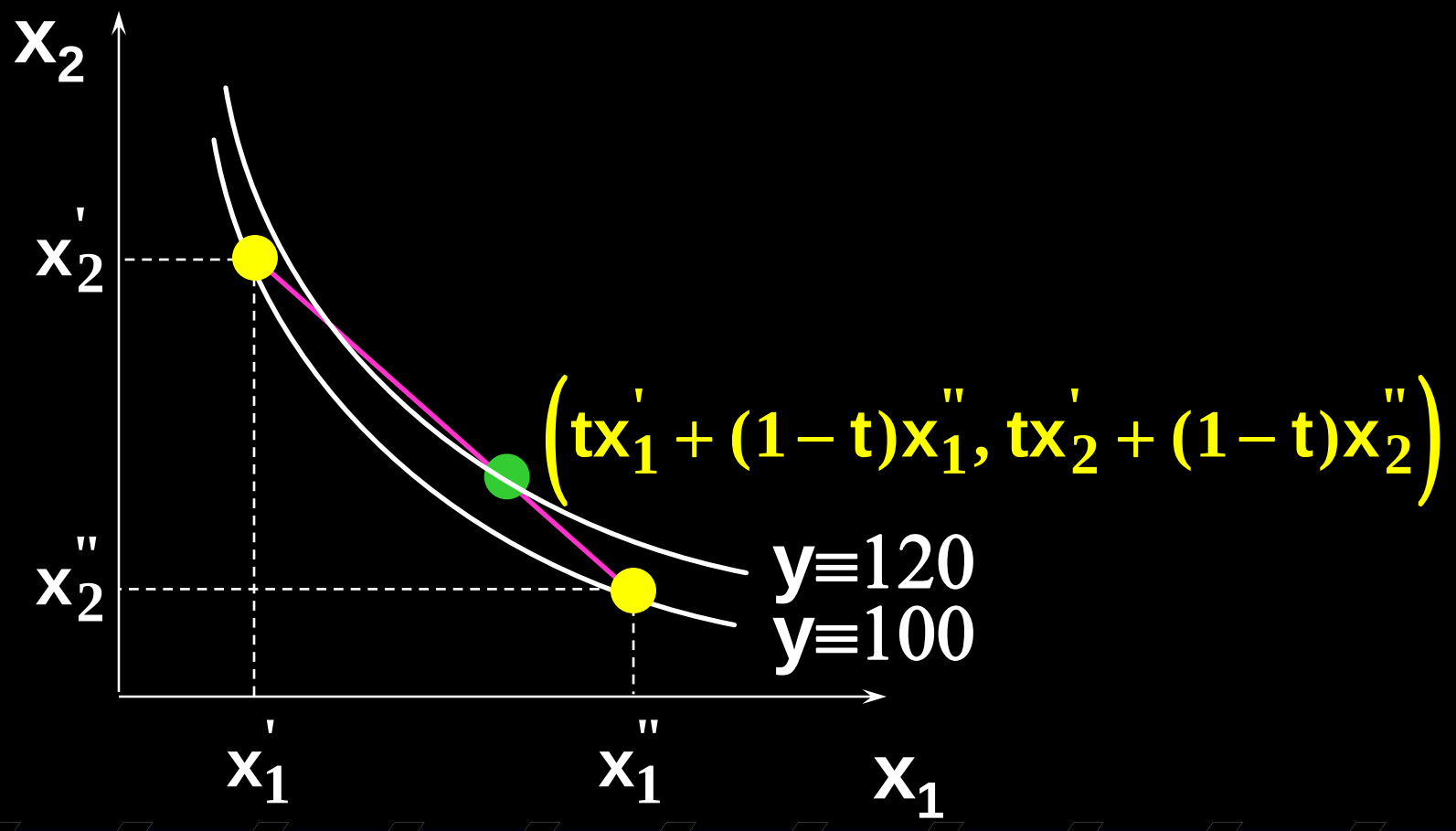
Well-Behaved Technologies - Convexity



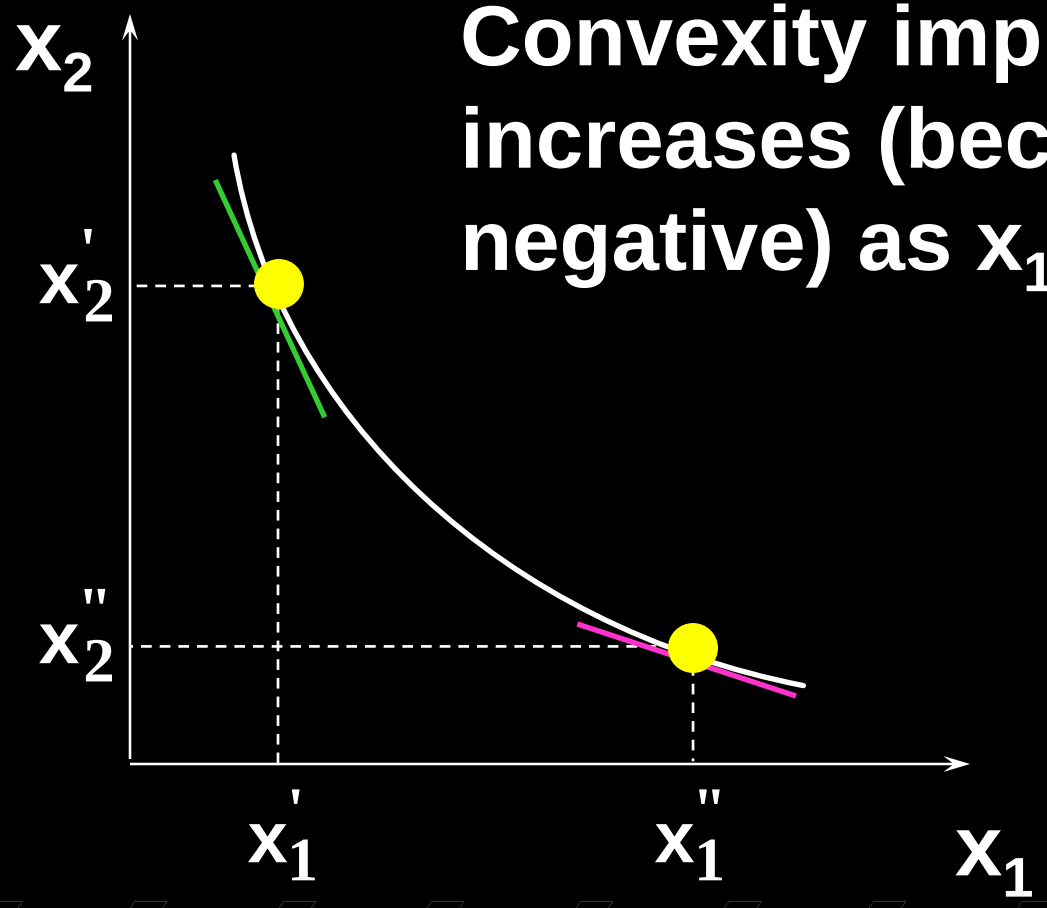
Well-Behaved Technologies - Convexity

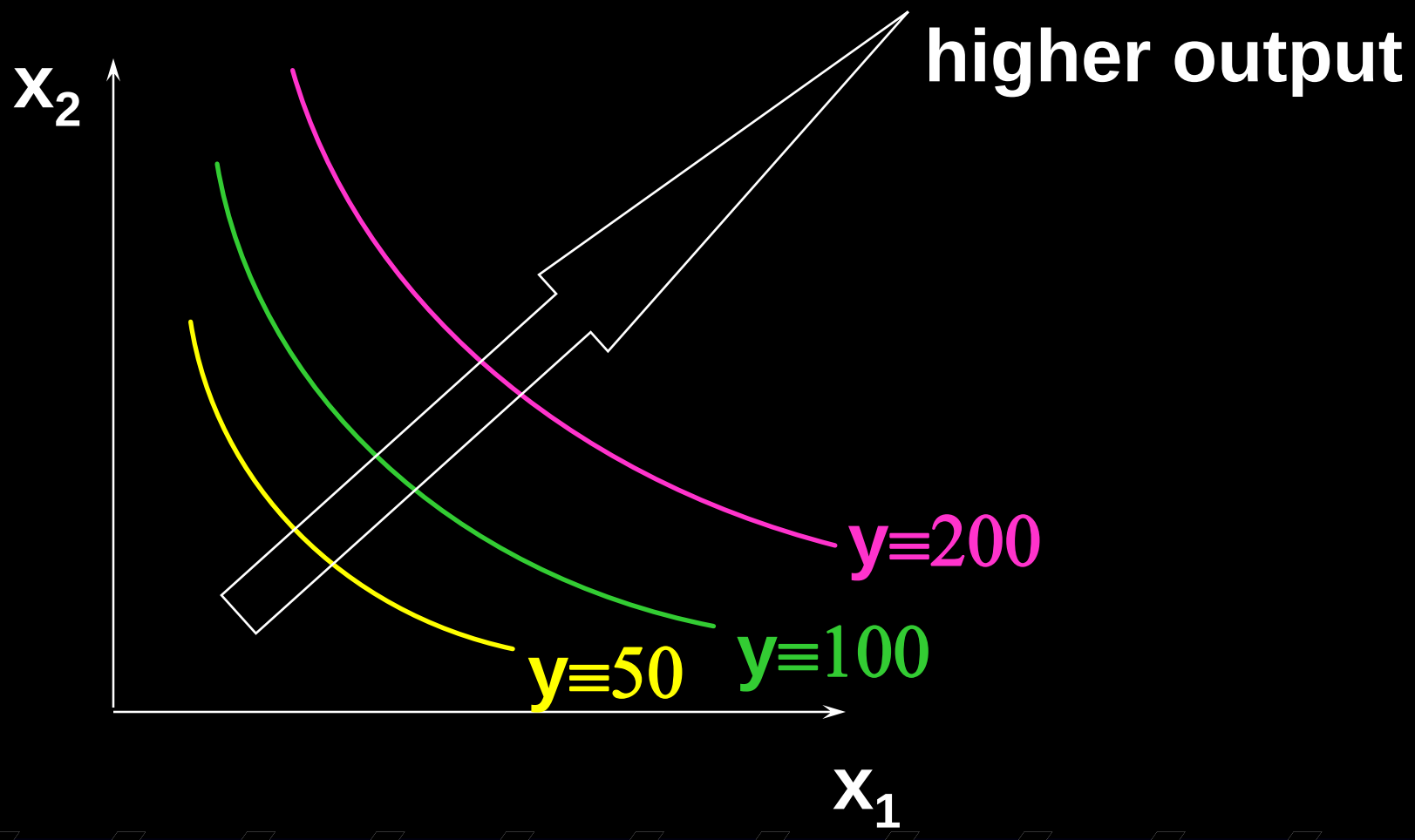


Well-Behaved Technologies - Convexity



Well-Behaved Technologies - Convexity



Convexity implies that the TRS increases (becomes less negative) as x_1 increases.

Well-Behaved Technologies